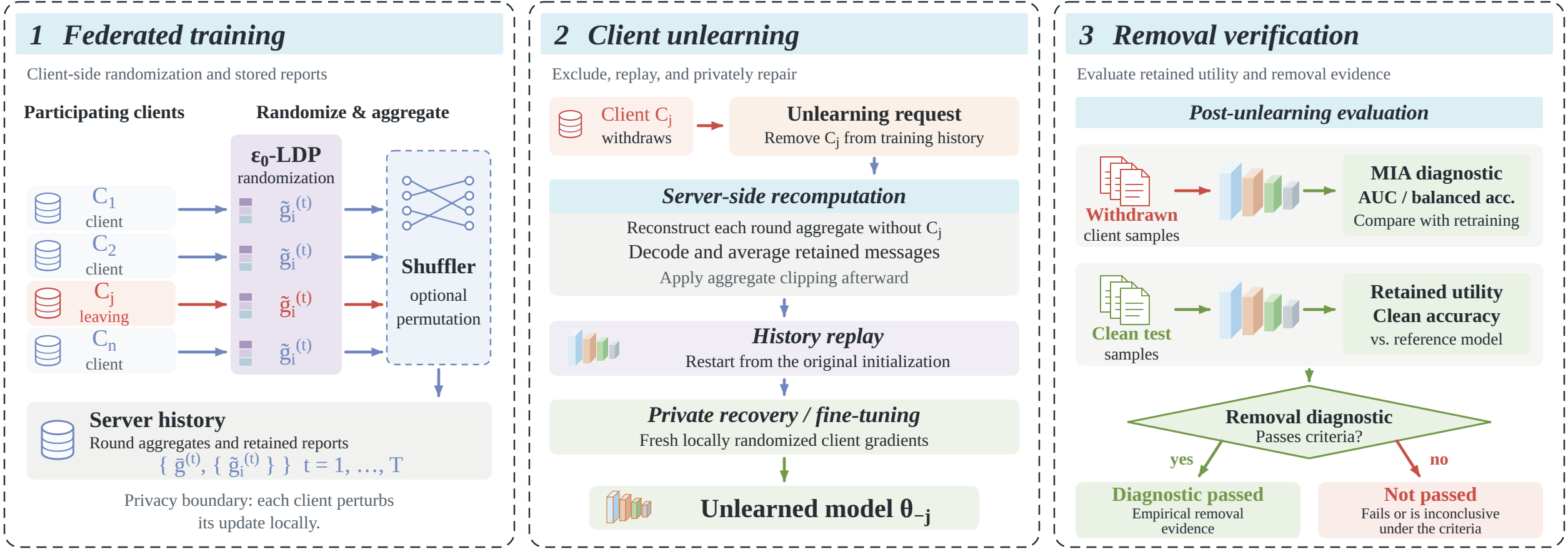
1 Federated training
Client-side randomization and stored reports
Participating clients
Randomize & aggregate
C1 client
C2 client
Cj leaving
Cn client
ε0-LDP randomization
Shuffler optional permutation
Server history
Round aggregates and retained reports
{ g̃(t), { g̃i(t) } } t = 1, …, T
Privacy boundary: each client perturbs its update locally.
2 Client unlearning
Exclude, replay, and privately repair
Client Cj withdraws
Unlearning request
Remove Cj from training history
Server-side recomputation
Reconstruct each round aggregate without Cj
Decode and average retained messages
Apply aggregate clipping afterward
History replay
Restart from the original initialization
Private recovery / fine-tuning
Fresh locally randomized client gradients
Unlearned model θ−j
3 Removal verification
Evaluate retained utility and removal evidence
Post-unlearning evaluation
Withdrawn client samples
MIA diagnostic
AUC / balanced acc.
Compare with retraining
Clean test samples
Retained utility
Clean accuracy
vs. reference model
Removal diagnostic
Passes criteria?
yes
no
Diagnostic passed
Empirical removal evidence
Not passed
Fails or is inconclusive under the criteria